

PRE-BOARD EXAMINATION (2024-25)
CLASS X

TIME: 3 HOURS

SUBJECT-SCIENCE (086)

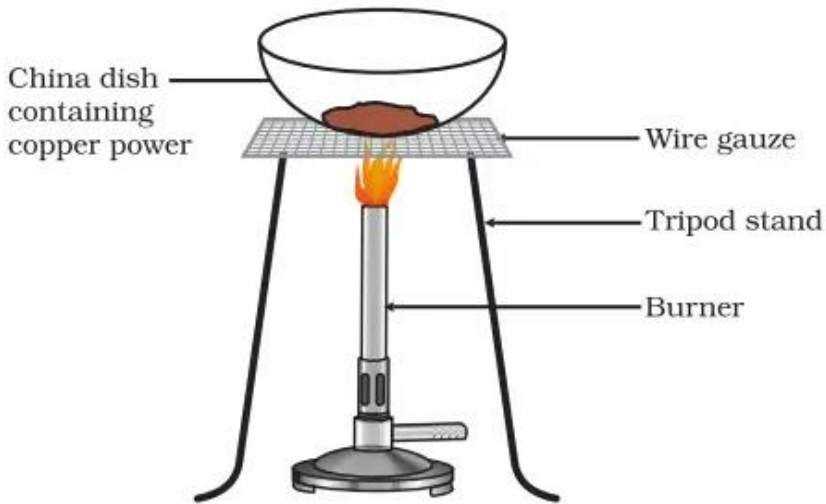
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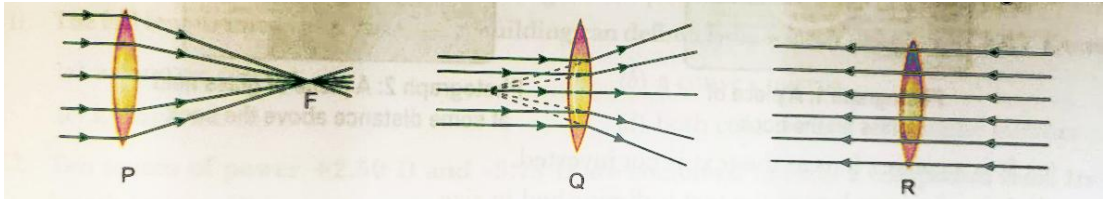
General Instructions:

1. All questions would be compulsory. However, an internal choice of approximately 33% would be provided. 50% marks are to be allotted to competency-based questions.
2. Section A would have 16 simple/complex MCQs and 04 Assertion-Reasoning type questions carrying 1 mark each.
3. Section B would have 6 Short Answer (SA) type questions carrying 02 marks each.
4. Section C would have 7 Short Answer (SA) type questions carrying 03 marks each.
5. Section D would have 3 Long Answer (LA) type questions carrying 05 marks each.
6. Section E would have 3 source based/case based/passage based/integrated units of assessment (04 marks each) with sub-parts of the values of 1/2/3 marks.

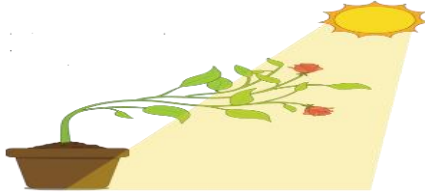
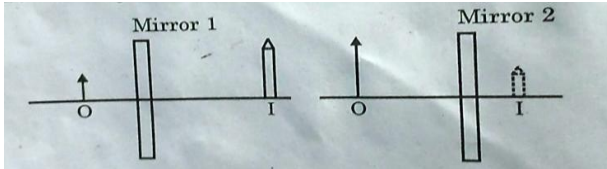
SECTION-A

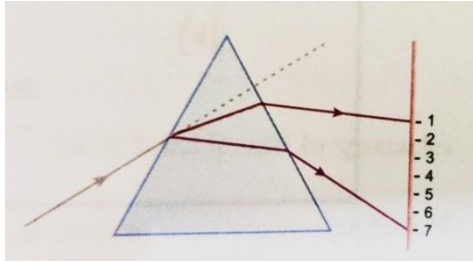
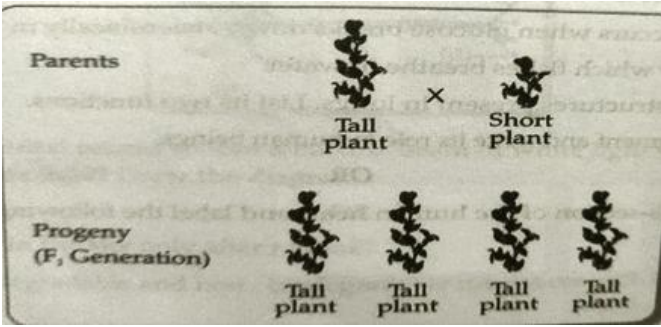
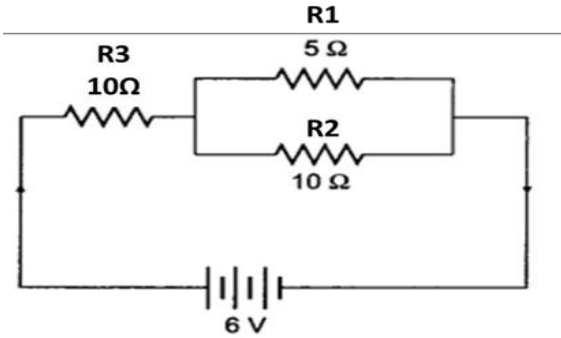
Select and write one most appropriate option out of the four options given for each of the question 1-20.

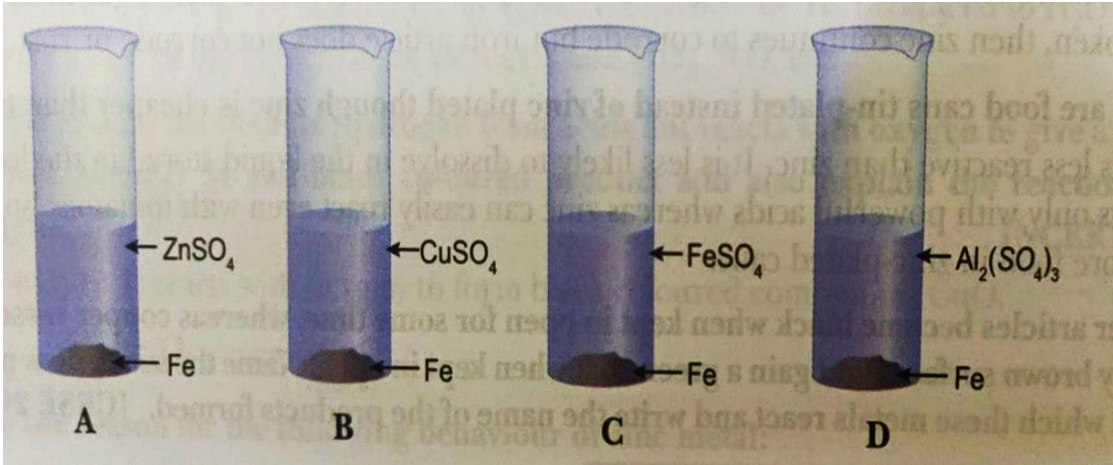
1	Which of the following belongs to a homologous series of alkynes? (A) C_6H_6 (B) C_2H_6 (C) C_2H_4 (D) C_3H_4	1
2	$MnO_2 + xHCl \longrightarrow MnCl_2 + yH_2O + zCl_2$ (A) 6,2,2 (B) 4,1,2 (C) 4,2,1 (D) 2,2,1	1
3	Right hand thumb rule is used to determine the: (A) Direction of magnetic field (B) Direction of magnetic force (C) Strength of magnetic field (D) Direction of induced current	1
4	Which of the following is correct for the given experimental diagram?  China dish containing copper powder Wire gauze Tripod stand Burner (A) Reddish brown coloured copper is reduced to black coloured copper(I)oxide (B) Reddish brown coloured copper is oxidized to black coloured copper(I)oxide (C) Reddish brown coloured copper is reduced to black coloured copper (II)oxide (D) Reddish brown coloured copper is oxidized to black coloured copper (II)oxide	1
5	Inside the magnet, the field lines move— (A) from north to south (B) from south to north (C) away from south pole (D) away from north pole	1

6	A plant gets rid of excess water through transpiration. What is the method used by plants to get rid of solid waste products? (A) Shortening of stem (B) Dropping down fruits (C) Shedding of yellow leaves (D) Expansion of roots into the soil	1
7	During the formation of a rainbow the position of observer and sun is (A) observer behind the sun (B) the sun behind the observer (C) observer facing the sun (D) at any position	1
8	Chromosomes: (i) carry hereditary information from parents to the next generation. (ii) are thread-like structures located inside the nucleus of an animal cell. (iii) always exist in pairs in human gametes. (iv) are involved in the process of cell division. The correct statements are : (A) (i) and (ii) (B) (iii) and (iv) (C) (i), (ii) and (iv) (D) (i) and (iv)	1
9	A few drops of turmeric solution are added to a colourless liquid. If the liquid becomes red, the liquid may be: (A) Hydrochloric acid (B) Distilled water (C) Ammonium hydroxide (D) Lemon juice	1
10	In the structure of cyclohexane (C_6H_{12}) there are: (A) Six single bonds, four double bonds and two triple bonds (B) Six single bonds, two double bonds and one triple bond (C) Eight single bonds and five double bonds (D) Eighteen single bonds only	1
11	A sportsman, after a long break of his routine exercise, suffered muscular cramps during a heavy exercise session. This happened due to: A) lack of carbon dioxide and formation of pyruvate. B) presence of oxygen and formation of ethanol. C) lack of oxygen and formation of lactic acid. D) lack of oxygen and formation of carbon dioxide.	1
12	Shown below is a photograph of a convex lens. A bright spot is seen on the paper when the lens is kept out facing the sun.  Which diagram below explain the formation of the bright spot?  (A) Only P (B) Only Q (C) only R (D) both P and Q	1
13	Which of the following features relates to biodegradable substances? A) Broken down by biological processes (B) Remain inert C) Persist in environment for long time (D) May harm the ecosystem	1

14	Common salt, besides being used in kitchen, can be used as the raw, material for making (i) Washing soda (iii) bleaching powder (ii) Baking soda (iv) slaked lime (A) (i) and (ii) (B) (i), (ii) and (iv) (C) (i) (ii) and (iii) (D) (i), (iii) and (iv)	1
15	Which of the following is a contraceptive device? A) Copper T B) Condom C) Diaphragm D) All of these	1
16	An element with atomic number _____ will form a basic oxide. (A) 7 (2,5) (B) 17 (2,8,7) (C) 14 (2,8,4) (D) 11 (2,8,1)	1
Question No. 17 to 20 consist of two statements – Assertion (A) and Reason (R). Answer these questions by selecting the appropriate option given below: (A) Both A and R are true, and R is the correct explanation of A. (B) Both A and R are true, and R is not the correct explanation of A. (C) A is true but R is false. (D) A is false but R is true		
17	Assertion (A): Insulin hormone is secreted by pituitary gland. Reason (R): If insulin is not secreted in proper amounts, the sugar level in the blood rises	1
18	Assertion(A) : On freely suspending a current-carrying solenoid, it comes to rest in N-S direction just like a bar magnet. Reason (R) : One end of current carrying straight solenoid behaves as a North pole and the other end as a South pole.	1
19	Assertion(A): The sex of the children will be determined by chromosome received from the father. Reason (R) : A human male has a pair of two X chromosomes.	1
20	Assertion(A) : On reacting with water, calcium starts floating over water. Reason(R): Calcium reacts with cold water at room temperature.	1
SECTION-B Q. no. 21 to 26 are very short answer questions: (VSA)		
21	Bile juice does not contain any digestive enzyme but still plays a significant role in the process of digestion. Justify the statement.	2
22	<u>Attempt either option A or B</u> A. Draw a food chain of at least three trophic levels. What will be the impact of removing organisms of second trophic level? Write two impacts. OR B. “Ozone, although a deadly poison, performs an essential function for the life on the earth.” Justify this statement.	2
23	<u>Attempt either option A or B</u> A. What will happen if ‘X’ is alcohol and ‘Y’ is carboxylic acid containing two carbons, and X and Y react in the presence of an acid catalyst. Name the reaction. Write chemical equation for the reaction. Name the product formed in the reaction. OR B. Surbhi and Ajay are studying about saturated and unsaturated hydrocarbons. Their teacher asks them which hydrocarbon can undergo substitution reaction? Surbhi answers that substitution reaction occurs in saturated Hydrocarbons, on the contrary Ajay answers, it occurs in unsaturated Hydrocarbons. Whose answer is correct? Justify with valid reason.	2

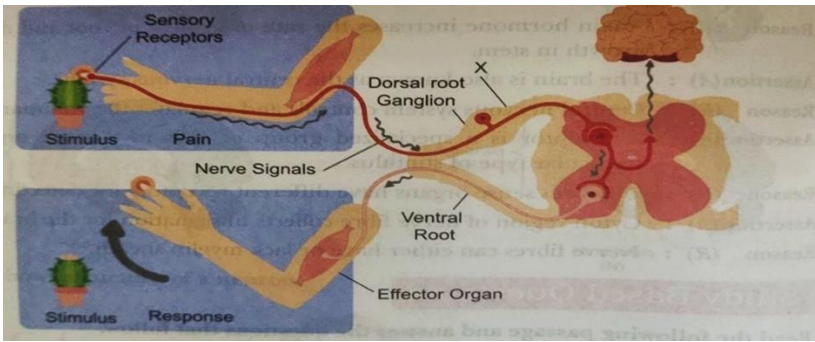
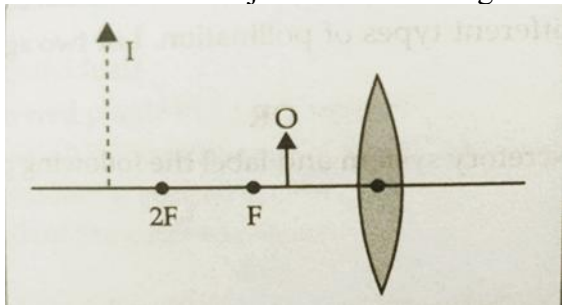
24	<u>Attempt either option A or B</u> A. Series arrangement of resistors is not used for domestic circuits? Give any two reasons for your answer. OR B. (i) How is the type of current used in household supply different from the current given by a battery of dry cells? (ii) Which one of the two currents mentioned above is considered to be more advantageous for the long-distance transmission of electric power and why?	2
25	Observe the image given below carefully and explain how this response is different from the response shown by leaves of touch me-not plant. 	2
26	Why reuse of material better than recycling? Write any two reasons.	2
SECTION-C Q. No. 27 to 33 are short answer questions (SA)		
27	<u>Attempt either option A or B</u> A. Take 3 g of barium hydroxide in a test tube, now add about 2 g of ammonium chloride and mix the contents with the help of a glass rod. Now touch the test tube from outside. (i) What do you feel on touching the test tube? (ii) State the inference about the type of reaction occurred. (iii) Write the balanced chemical equation of the reaction involved OR B. In the electrolysis of acidulated water (i) Name the gas collected at the (a) cathode and (b) anode. (ii) Why is the volume of one gas collected at one electrode two times the gas collected at the other electrode? (iii) Write the method of testing of any one of the gases evolved in this case.	3
28	<u>Attempt either option A or B</u> A. (i) How do guard cells regulate opening and closing of stomatal pores? (ii) Leaves of healthy potted plant were coated with vaseline. Will this plant remain healthy for long? Give reason for your answer OR B. (i) List any two characteristics of lungs which make them an efficient respiratory system. (ii) Name the part of human respiratory system: (a) which separates chest cavity from abdominal cavity. (b) which has balloon-like structures where exchange of gases takes place.	3
29	Study the following diagram:  (i) Identify the type of spherical mirror represented as Mirror 1 and Mirror 2 in the above figure and state the use of each. (ii) Define absolute refractive index.	3
30	A substance 'X' is used in building materials and is insoluble in water. When it reacts with dil HCl, it produces salt and a gas which turns lime water milky. (i) Write the chemical name and formula of 'X'. (ii) Write the chemical equations for the both chemical reactions involved in the above statement.	3

31	<u>Attempt either option A or B</u> A. A person needs a lens of power -4.5 D for correction of her vision. (i) What kind of defect in vision is she suffering from? (ii) What is the focal length of the corrective lens? (iii) What is the nature of corrective lens? OR B. A beam of white light falling on a glass prism gets split up into seven colours marked 1 to 7 as drawn in the diagram. A student makes following statements about the spectrum observed on the screen.  (i) The colours at position 3 and 5 are similar to the colour of the sky and the colour of gold metal respectively. Is the statement made by the student correct or incorrect? Justify (ii) Which two positions correspond closely to the colour of: (a) Brinjal (b) 'danger' or stop signal lights	3
32	Sahil performed an experiment to study the inheritance pattern of genes. He crossed tall pea plant 'TT' with short pea plants 'tt' and obtained all tall pea plants in F₁ generation.  (i) What will be the genotype of plants in the F₁ generation? (ii) Give reason why only tall plants are observed in F₁ progeny? (iii) When F₁ plants were self-pollinated, a total of 800 plants were produced. How many of these would be tall and short plants?	3
33	<u>Attempt either option A or B</u> A. (i) What is a solenoid? Draw the magnetic lines of force around a solenoid. (ii) What is the effect of placing an iron core in the side the solenoid? OR B. (i) Why is it necessary to earth the metallic appliances? (ii) How is fuse connected in domestic circuit and how does an electric fuse act as a safety device?	
SECTION-D Q.no.34 to 36 are long answer questions (LA)		
34	<u>Attempt either option A or B</u> A. Observe the given line diagram of electric circuit and answer the following questions. 	5

	(i) What is the total resistance of R1 and R2 only. (ii) What will be total resistance if all the resistance readjusts to join in a series form. (iii) Calculate the total resistance of the circuit if resistance R1 and R3 are exchanged to each other in the given circuit diagram. (iv) State ohm's law. OR B. (i) List two factors on which the resistance of a conductor in the shape of a straight cylinder depends. Write the relation between resistance and the resistivity of the material of the conductor. Use this relation to obtain the SI unit of resistivity. (ii) Resistance of a metal wire of length 1 m is $40\ \Omega$. If the area of cross-section of the wire is $6.5 \times 10^{-8}\text{ m}^2$, find the resistivity of the material of the wire	
35	<u>Attempt either option A or B</u> A. (i) Certain specialized cells in animals called stem cells have the ability to divide and differentiate into different cell types. This helps in replacement of a damaged organ. Name and explain one method of asexual reproduction, if the individual is somehow broken up into pieces, each of its pieces can grow into separate individuals. (ii) Identify two pairs of reproductive organs in males and females that are functionally similar to each other. Justify? OR B. (i) "Use of a condom is beneficial for both sexes involved in a sexual act." Justify this statement giving two reasons. (ii) How is oral contraceptive pills help in avoiding pregnancies? (iii) What is sex selective abortion? How does it affect a healthy society? (any one consequence)	2+3 2+1+2
36	<u>Attempt either option A or B</u> A. Sakshi was comparing the activity of different metals for her science project. She added iron filings in four test tubes A,B,C, D containing solutions ZnSO_4, CuSO_4, FeSO_4 and $\text{Al}_2(\text{SO}_4)_3$ respectively as shown in the figure:  (i) In which of the test tubes she will observe the reaction to be most vigorous. (ii) What is the reason for her observation. (iii) Write a well-balanced equation of the reaction for the above case (iv) By transfer of electrons, illustrate the formation of bond in magnesium chloride and identify the ions present in this compound. OR B. (i) Write the name of two metals of moderate reactivity. How can this be demonstrated experimentally in a school laboratory that which one of the two is more reactive? (ii) Some metals of moderate reactivity are also found in the Earth's crust in the form of their carbonates. Name the process involved in the extraction of metal from them, giving chemical equations for the reactions that occur.	5

SECTION-E

Q.no.37 to 39 are case-based / data-based questions.

Q37	Rajesh accidentally touched a thorn but quickly withdrew his hand. He later realized that he did this without even thinking about it! So, his reflexes were quite quick.  (A) What is the main center for such reflex actions of Rajesh? (B) In a neuron, the conversion of electrical signal to a chemical signal occurs at which part? <u>Attempt either option C or D</u> (C) What are electrical impulses? Write any two limitations of electrical impulses. OR (D) Write the sequence of events that occurs in reflex action.	4
Q38	Ethanol (commonly known as alcohol) is an important carbon compound widely used in industries, hospitals, etc. Because it is a very good solvent, it is used in medicines such as cough syrups, tincture iodine and many tonics. As it is a cleaner fuel, ethanol is used as an additive in petrol in most of the developed countries of the world. (A) Write the structure of ethanol. (B) Name the products formed when ethanol burns in sufficient air (oxygen). <u>Attempt either option C or D</u> (C) What happens when 5% solution of alkaline potassium permanganate is added to warm ethanol. Write the chemical equation. Name the product formed and state the role of 5% solution of alkaline potassium permanganate in the reaction. OR (D) What happens when ethanol is heated with excess conc. H_2SO_4 at 443 K. Write the chemical equation. Name the product formed and state the role of conc. H_2SO_4 in the reaction.	4
Q39	The diagram given below shows an object O and its Image I.  (A) Type of lens (B) Name an optical instrument where such an image is obtained. <u>Attempt either option C or D</u> (C) Draw the ray diagram to show image formation in the above shown case. OR (D) List two characteristic of image formed if this lens is replaced by a concave mirror of focal length f and an object placed at a distance $f/2$ in front of the mirror.	4